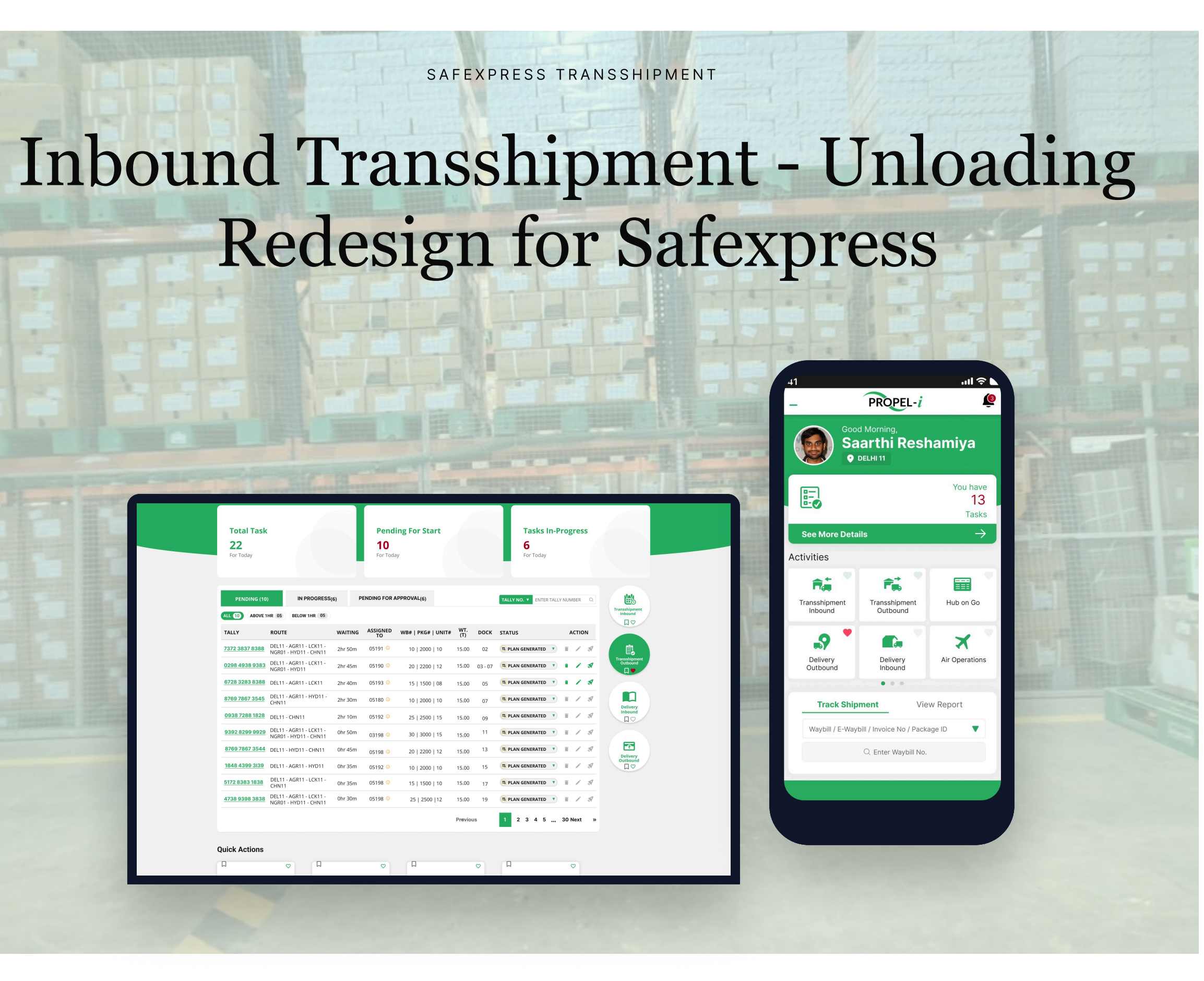
SAFEXPRESS TRANSSHIPMENT

Inbound Transshipment - Unloading Redesign for Safexpress


Total Task
22
Pending for Start
10
Tasks In Progress
6
Quick Actions
PROPEL-I
Saarthi Reshamiya
13
Track Shipment
View Report

Safexpress operates a network of regional hubs; the inbound transshipment segment handles truck arrival, unloading, sorting and onward dispatch. **Safexpress's inbound transshipment process was clogged at the unloading stage — manual workflows, weak visibility for team leads, high error rates in scanning, thereby delaying downstream dispatch.**

Over the course of several weeks, I led the design effort to discover and document warehouse workflows, create intuitive mobile interfaces, conduct stakeholder workshops, and guide the team through an agile design process.

We successfully delivered a fully functional unloading management system that is currently being piloted in several warehouses and set the foundation for additional features for the system to be scaled up.

Expertise:

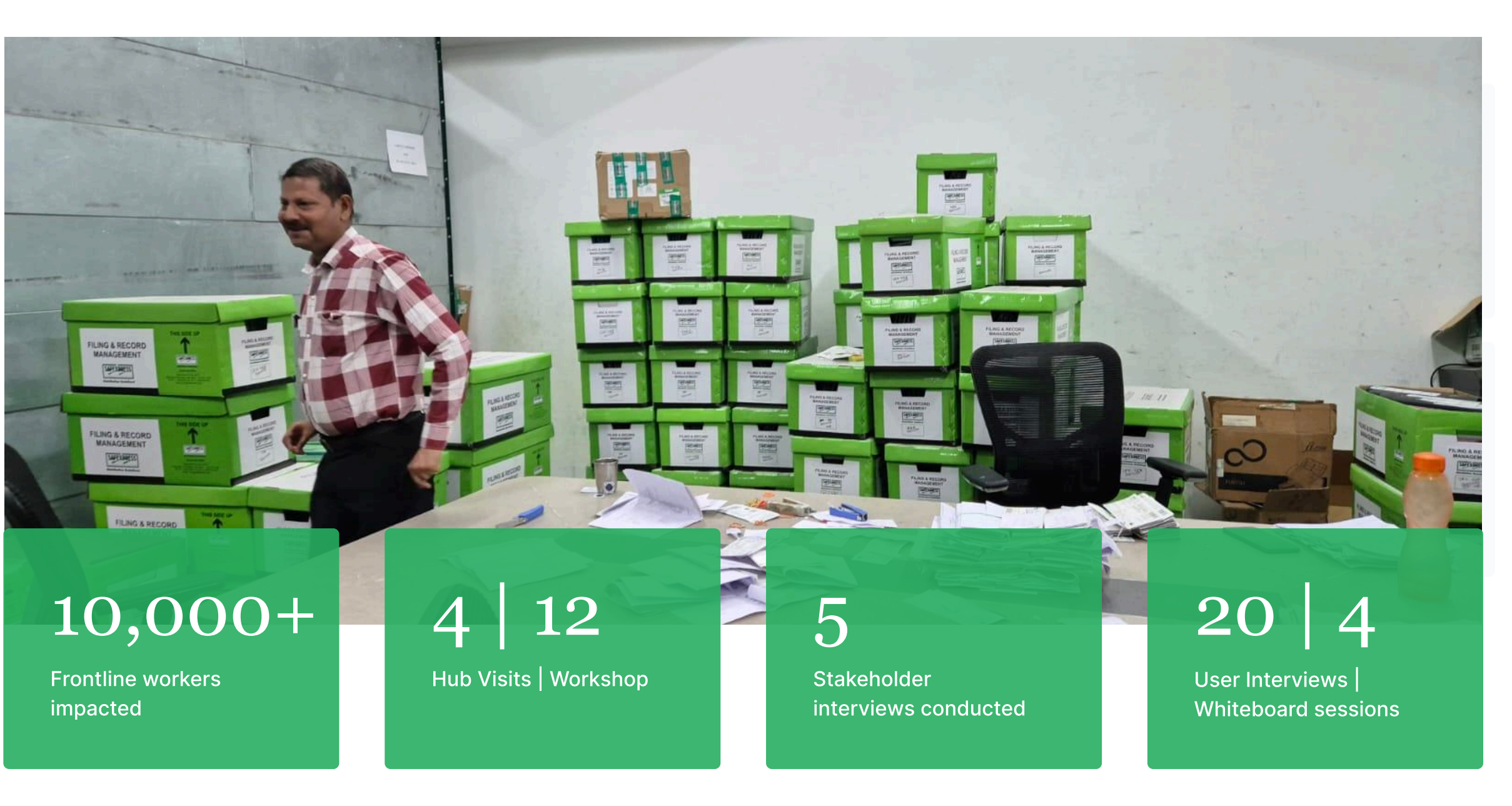
- UX design and research
- Mobile-first design for frontline workers
- Workflow optimization and process mapping

Project Details:

- Client: Safexpress
- Role: Senior UX Designer
- Time: 2024

Key Collaborators:

- Stakeholders
- Product Owner
- Operations Team
- Development Team



10,000+
Frontline workers impacted

4 | 12
Hub Visits | Workshop

5
Stakeholder interviews conducted

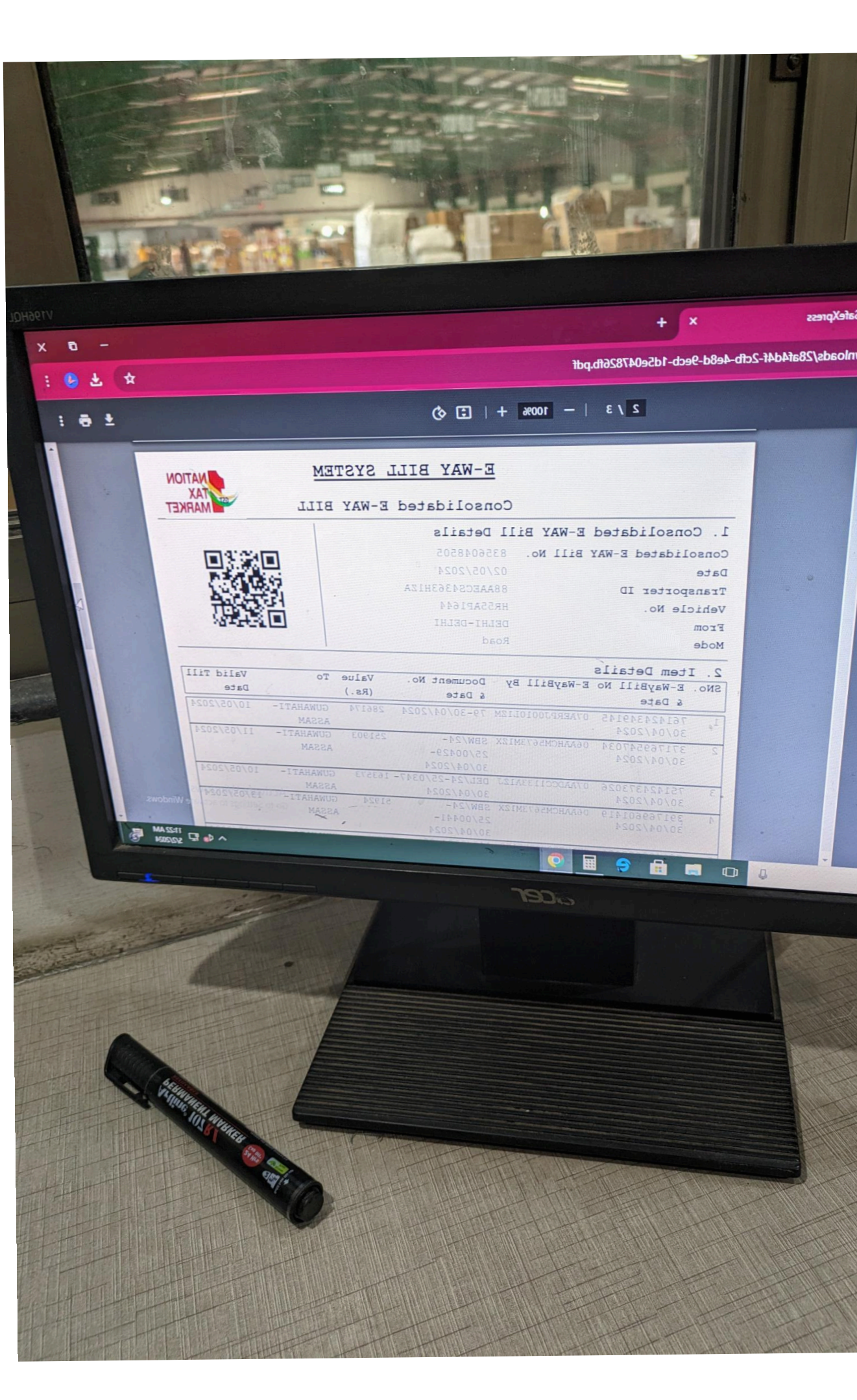
20 | 4
User Interviews | Whiteboard sessions

Challenge

Modernizing operations with minimal impact to familiar workflows

Unloading operations are currently challenged by inconsistent unloading times, limited real-time visibility, and reliance on manual coordination. Scanning mismatches and delayed system updates prevent supervisors from accurately tracking unloading progress, resulting in reactive decision-making and frequent team idle periods. The absence of live operational data makes it difficult to balance workloads, identify bottlenecks, or intervene promptly during delays.

Communication on the floor is largely manual, depending on verbal updates and phone calls, which leads to confusion, misalignment across teams, and avoidable downtime. In addition, reliability and usability issues with the handheld application disrupt workflows and reduce confidence in system data, further impacting productivity and data accuracy.



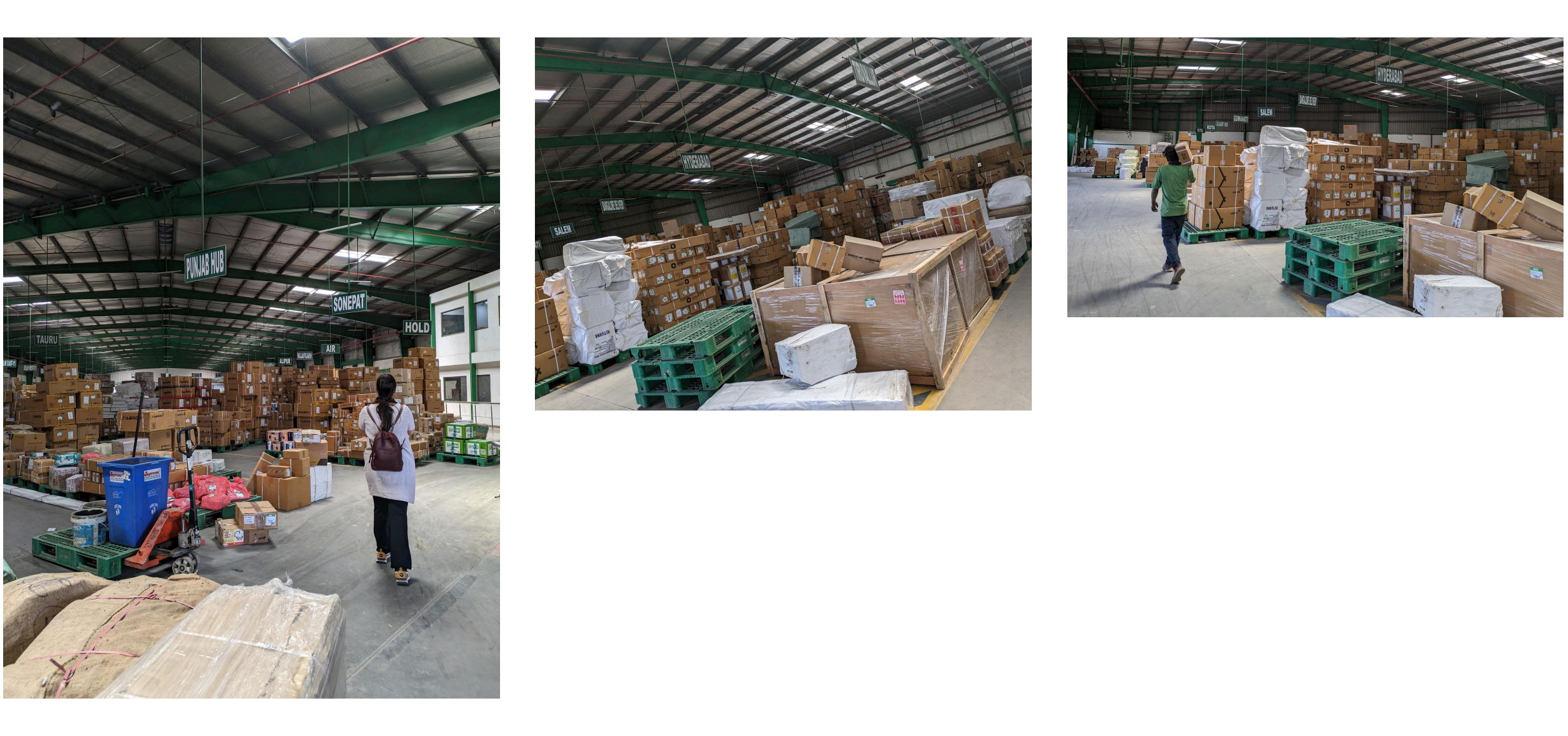
The lack of clear performance metrics and real-time dashboards limits effective performance monitoring and accountability across shifts. Overall, insufficient automation and digital transparency create operational inefficiencies, reduce labor utilization, and hinder consistent execution of unloading activities.

Approach

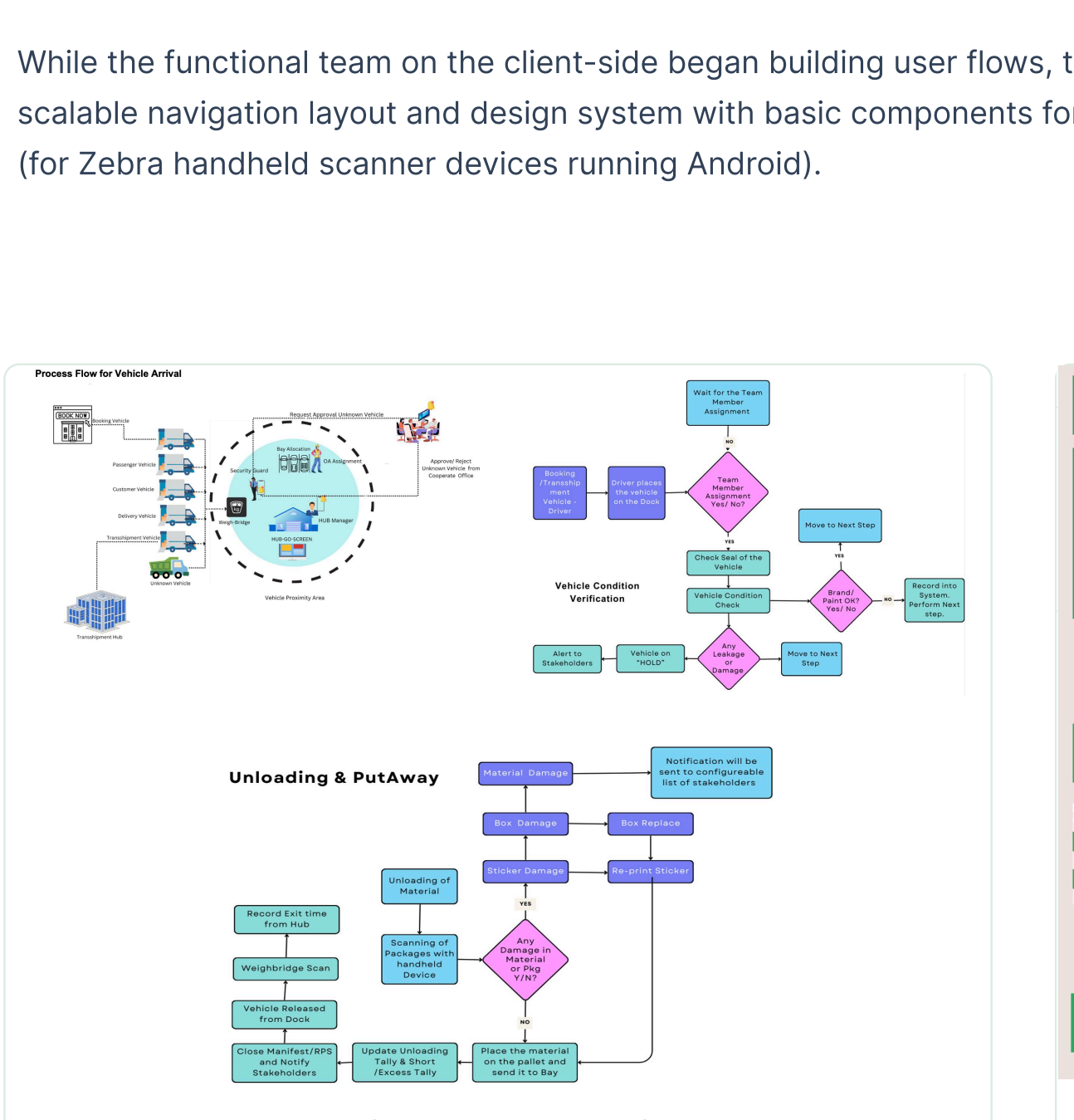
Ride-along research leads to empathy, and uncovering opportunities

To observe and document the workflows at the various functional areas, the team and I split up to different areas of the warehouse to shadow employees. Over the course of a few days of shadowing, we gathered dozens of photos, pages of notes, and videos. Most interesting to me were to various paper-based work-arounds that were created to fill gaps in the system.

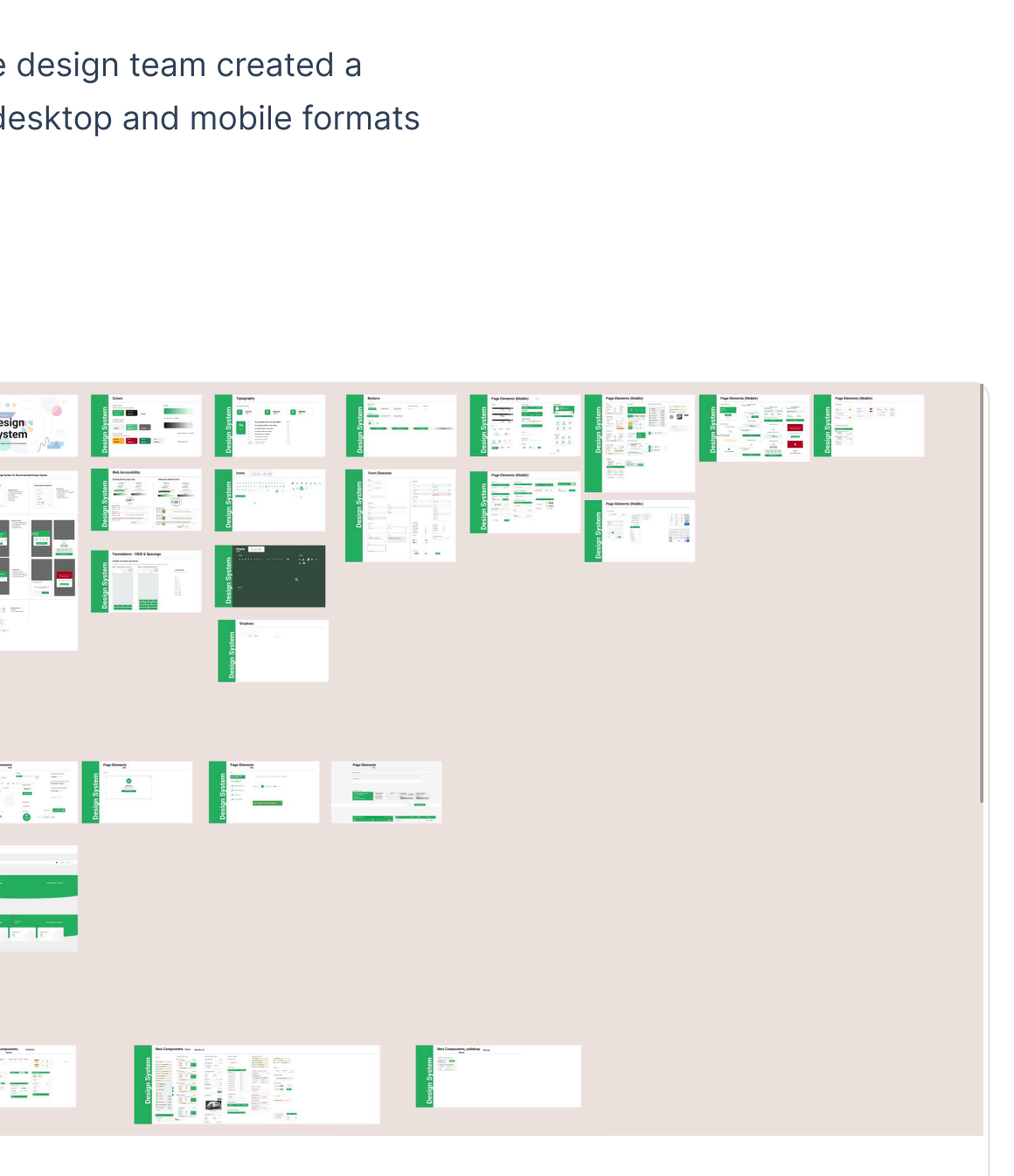
During the stakeholder workshop and Hub visit, the team provided a walkthrough, focusing on how goods are received and unloaded at the hub. The session helped visualize the complete operational flow—from arrival and dock allocation to unloading, sorting, and movement to outbound areas.



While the functional team on the client-side began building user flows, the design team created a scalable navigation layout and design system with basic components for desktop and mobile formats (for Zebra handheld scanner devices running Android).

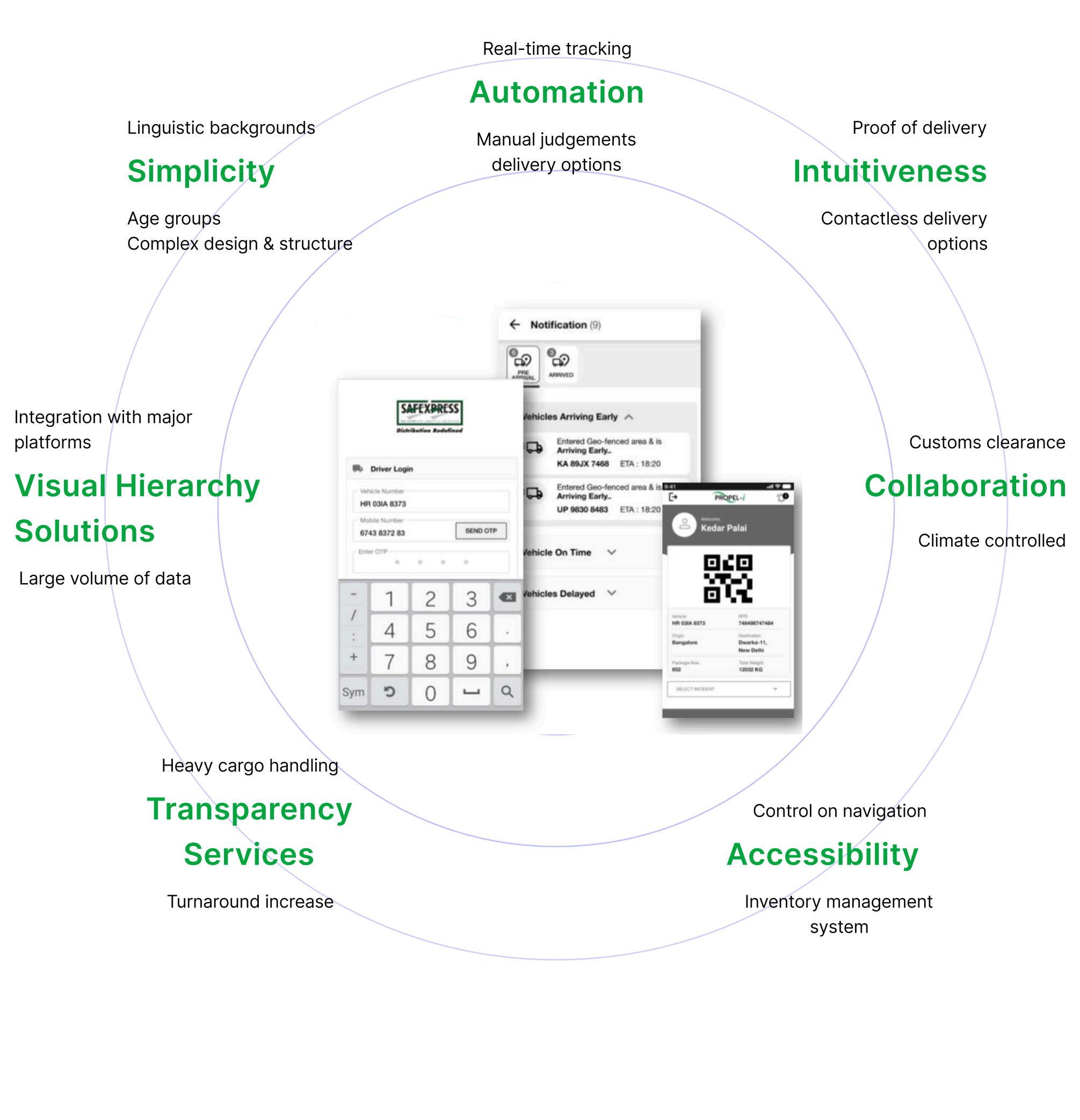

Process Flow for Vehicle Arrival
Unloading & PutAway

Few of many detailed user flows

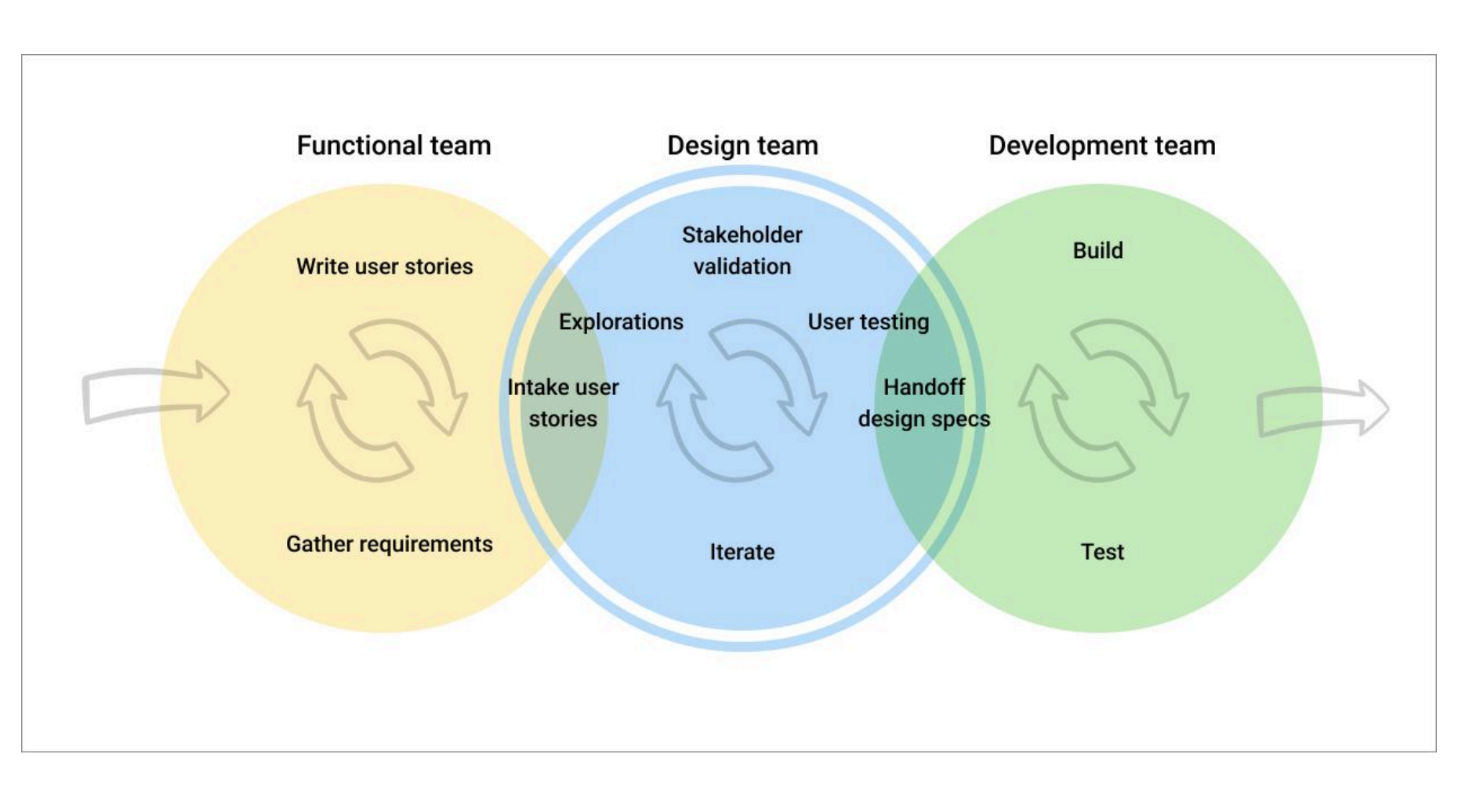


Design system page for actions

Soon, the team was creating wireframes. With the design system, we were able to quickly move into UI mock-ups.


Real-time tracking
Automation
Manual judgement delivery options
Linguistic backgrounds
Simplicity
Age groups
Complex design & structure
Proof of delivery
Intuitiveness
Contactless delivery options
Integration with major platforms
Visual Hierarchy Solutions
Large volume of data
Customs clearance
Collaboration
Climate controlled
Heavy cargo handling
Transparency Services
Turnaround increase
Control on navigation
Accessibility
Inventory management system

However, we began running into issues with the user stories. The sprint cycle kept on running into roadblocks with validating requirements, which led to the design team getting blocked. As it turned out, just observing employees and learning their workflows did not translate into robust, design-ready user stories. The client's functional team was new to Agile methodology, and needed more support to be successful, so we decided to take a pause to re-assess the project scope and needs.

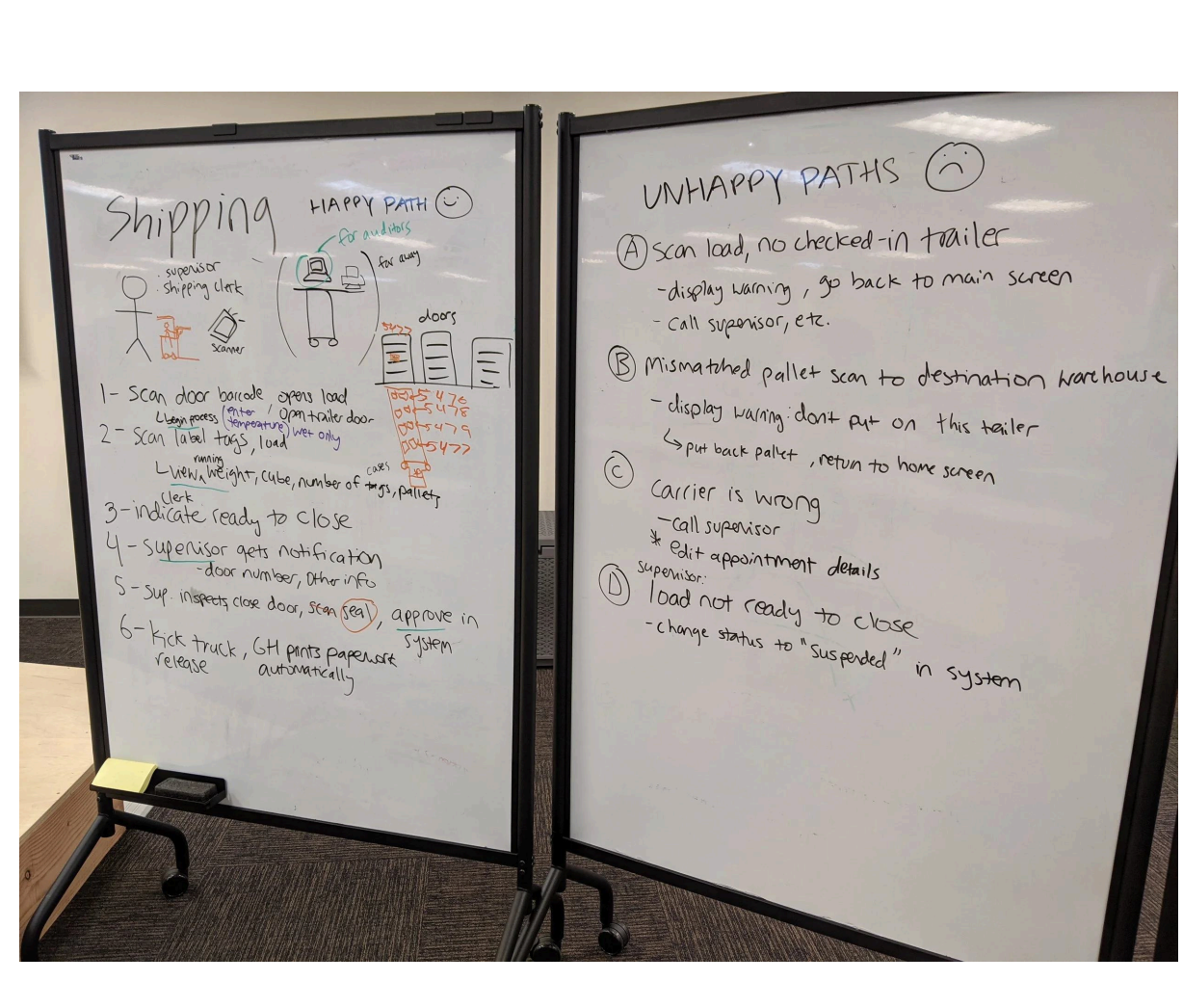

Functional team
Write user stories
Gather requirements
Design team
Stakeholder validation
Explorations
Intake user stories
Iterate
User testing
Handoff design specs
Development team
Build
Test

The typical design team workflow and how the teams collaborate

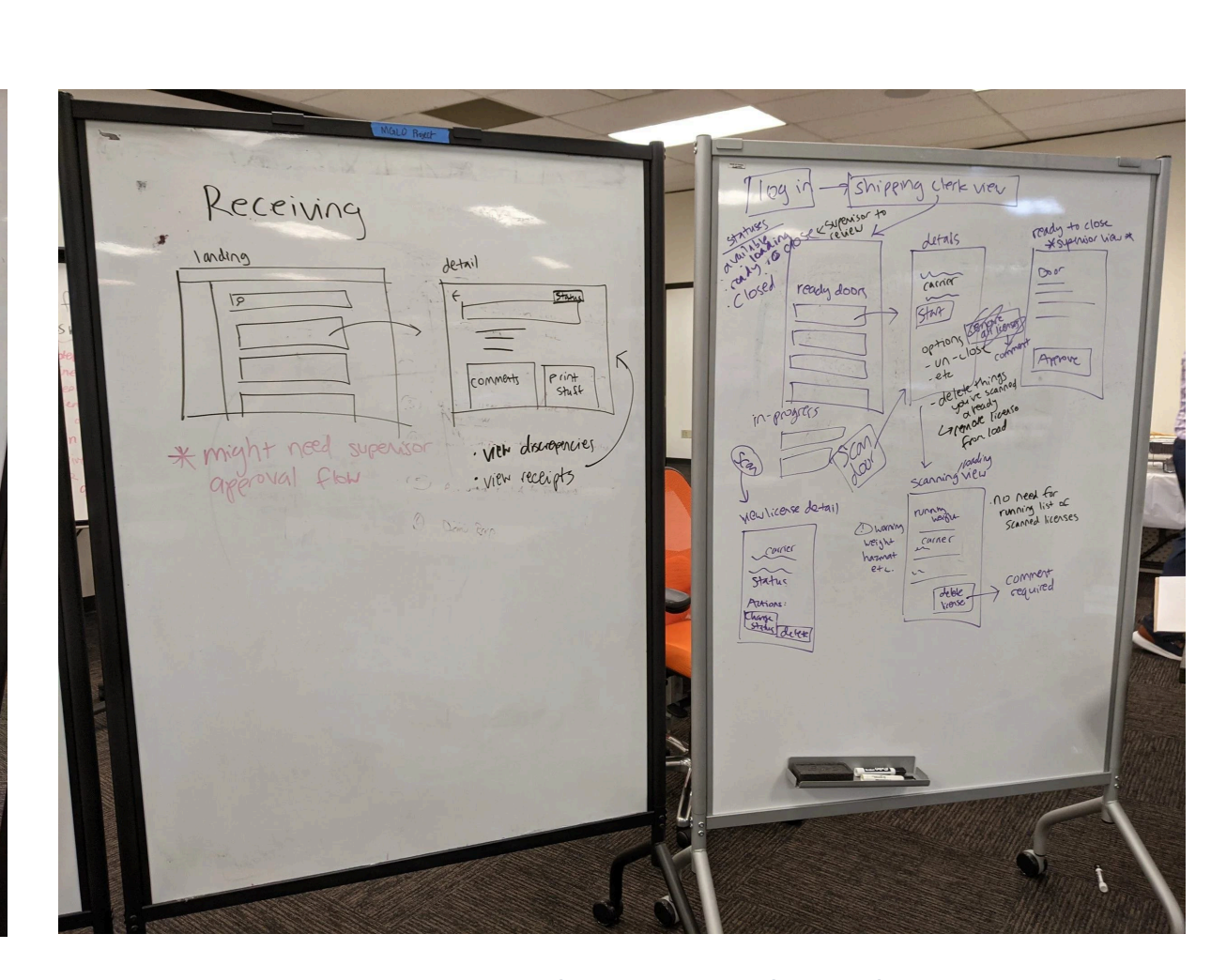
Re-Approach

Product visioning workshops pave the way for a clearer future

Over the course of a week, the project leadership and I led a series of visioning workshops to level-set on the scope of the project and deep-dive into each of the functional areas. Armed with several big whiteboards, I facilitated discussions with the client's functional team and subject-matter experts to define Happy Path and Unhappy Paths for each functional area. We then did high-level sketches of key screens. Having all this information made visible helped align all stakeholders and gave the design team more solid requirements to work from.



One of many whiteboards



Simple boxes define the vision for the flow

Since the week-long visioning workshops were not initially part of the project timeline, I also re-worked the design team timeline to account for the change in scope as well as an extension of the project.

With these workshops complete, I was able to more confidently educate the client's functional team on Agile best practices and help them write clear user stories. The design and dev teams could then efficiently iterate on designs and build features with fewer blockers.

Design

A versatile and robust system that follows UX best practices

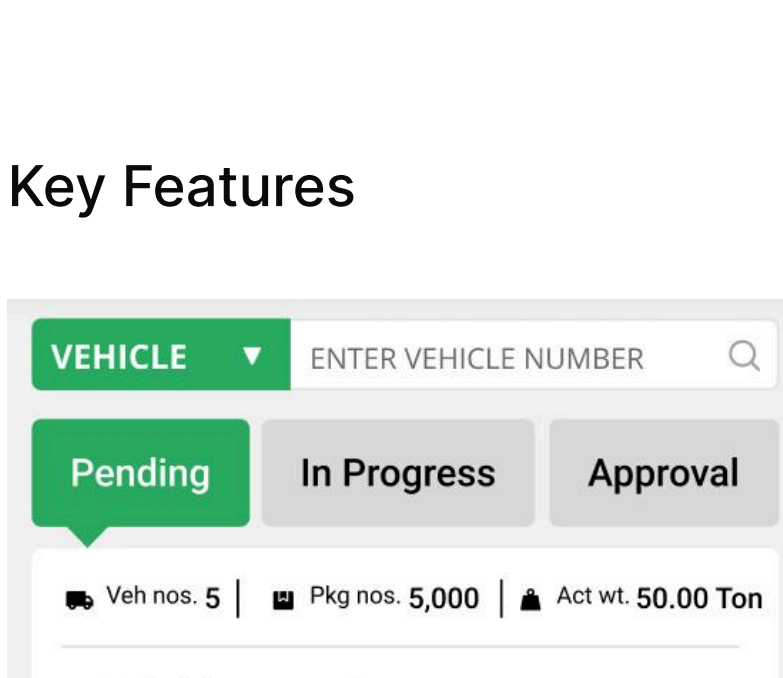
FEATURE HIGHLIGHTS

Unloading Dashboard

Everything that comes into view when team members start an unloading session. We designed these screens to make finding trucks, scanning waybills, and viewing to be easy and straightforward with a clear information hierarchy.

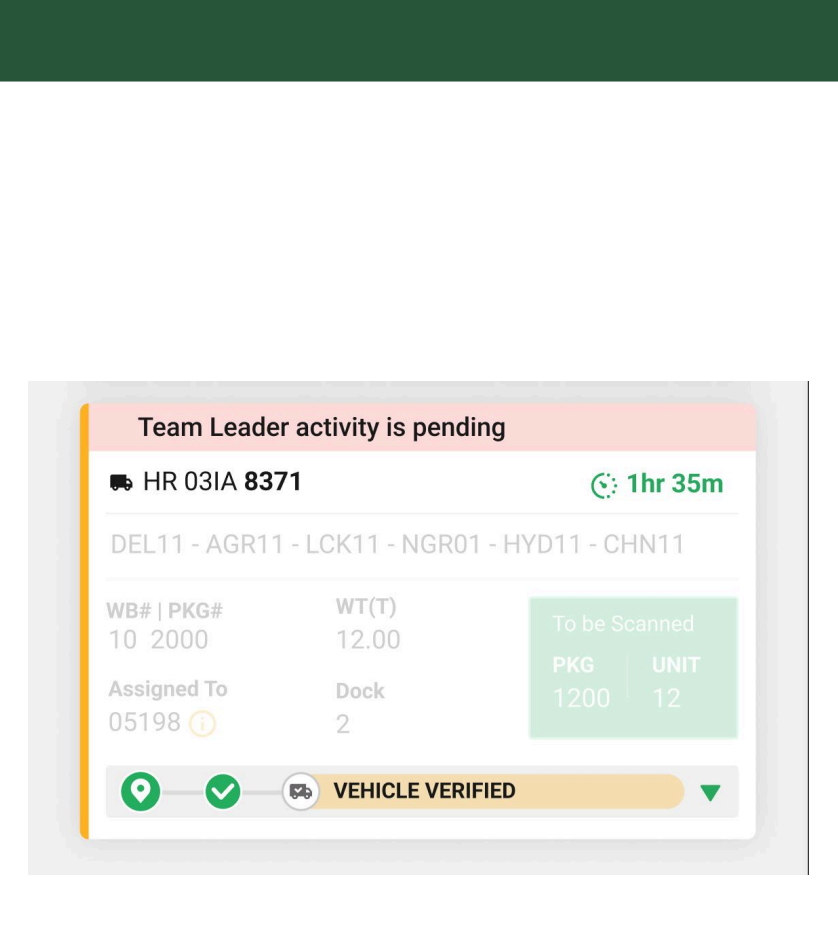
- Clear visual hierarchy for quick scanning
- Real-time status updates
- Optimized for handheld devices

Key Features


VEHICLE
ENTER VEHICLE NUMBER
Pending
In Progress
Approval
Vehicle(s) Reported
Above 1hr
Below 1hr

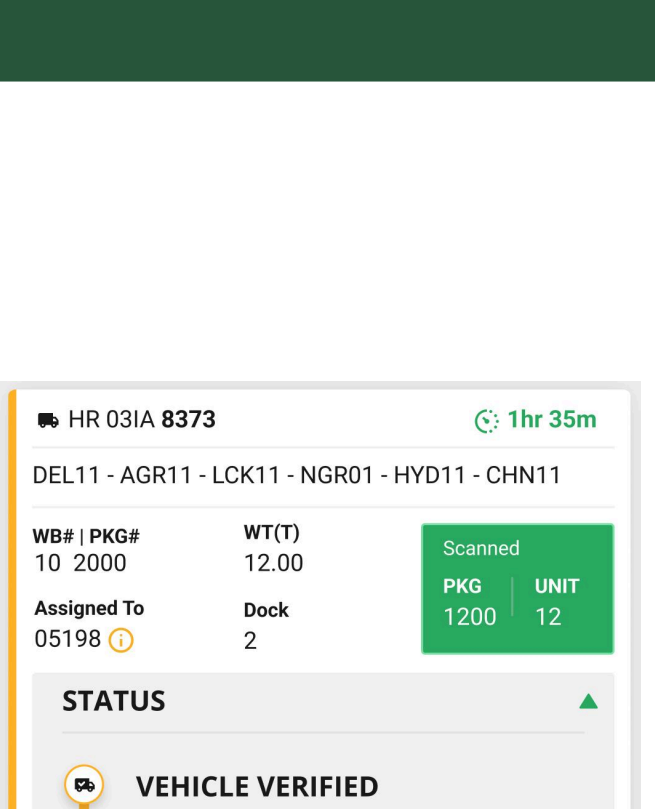
Category Search & Tabs

Team members see a clear list of trucks assigned to them for the shift with priority indicators.


Team Leader activity is pending
HR 03IA 8371
1hr 35m
VEHICLE VERIFIED

Quick Scanning

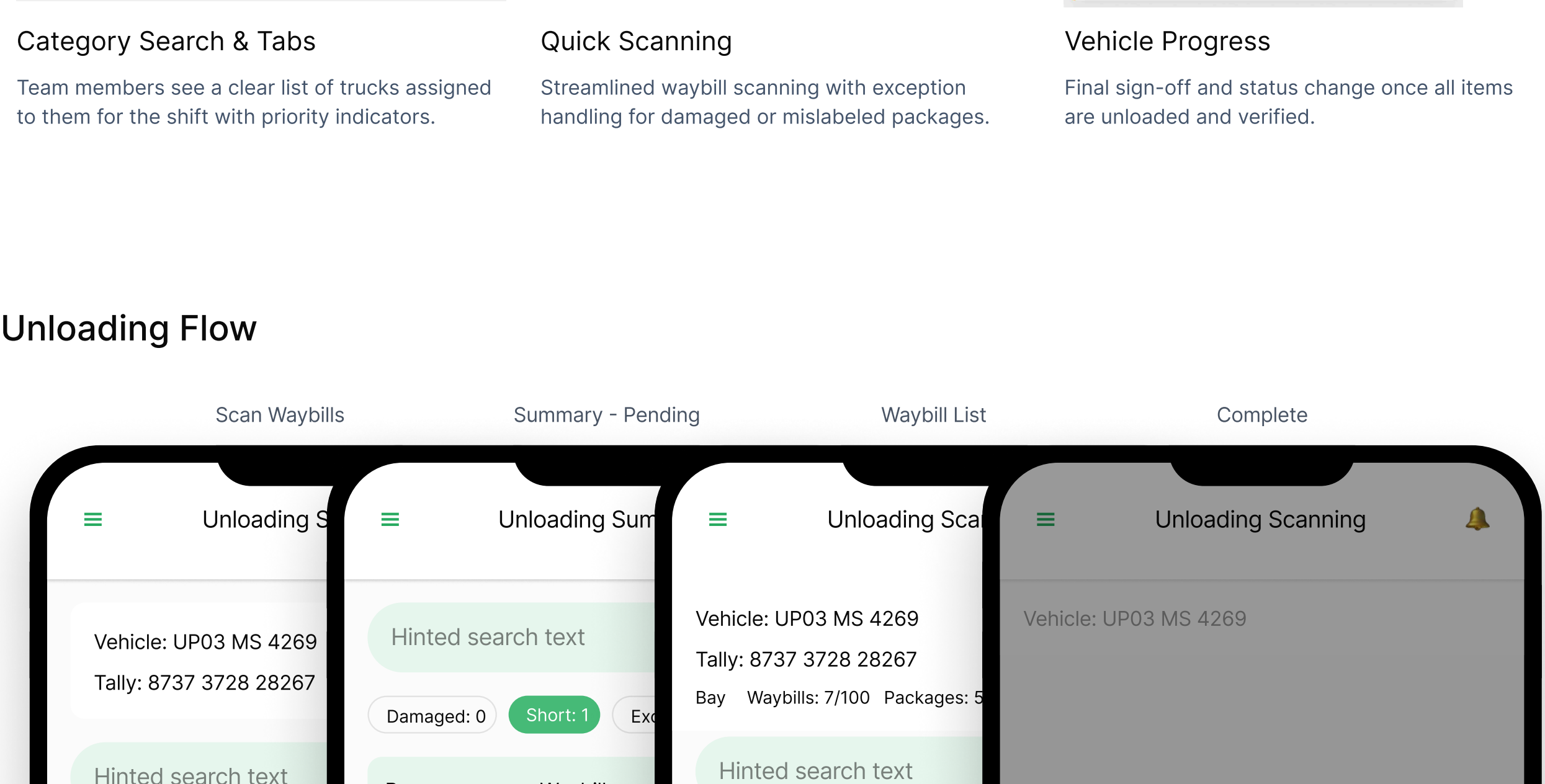
Streamlined waybill scanning with exception handling for damaged or mislabeled packages.


HR 03IA 8373
1hr 35m
STATUS
VEHICLE VERIFIED
REPORTED
ARRIVING

Vehicle Progress

Final sign-off and status change once all items are unloaded and verified.

Unloading Flow


Scan Waybills
Summary - Pending
Waybill List
Complete
Unloading Summary
Unloading Scanning
Vehicle: UP03 MS 4269
Hinted search text
Bay
Waybills
Bangalore
Gwalior
Gurugram
Mark as complete
Confirm
Next
Unloading Complete!
Sent to TL for approval
Start another task

Outcome

A fully functional system and an eager client team with new skills

With these workshops complete, I was able to more confidently educate the client's functional team on Agile best practices and help them write clear user stories. The design and dev teams could then efficiently iterate on designs and build features with fewer blockers.

Reflection

Addressing the elephant in the room can take up time, but saves more time in the future

When the team first decided to pause to conduct the product visioning workshops, I was worried about how much that would impact our timelines. The design team was already behind schedule on several features due to churn with the requirements. However, stopping to re-visit the scope and user journeys ended up being just what we need to move faster. The 1-week pause in designing led to much more clarity and greatly reduced the number of iterations needed for each design flow. Furthermore, the project leadership recognized the need for a good team and were on board with extending the overall project timeline, affording us the space and resources to put our best efforts into the work. Sometimes, the fastest way to move forward is to stop and take a look around first

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